

Supplementary Material

Inferring Institutionally Shaped Operational Constraints in Power System Dispatch Models

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Working revision: S1–S5 use the final 100-run datasets and the completed 2016 and 2021 original-UCED revalidation.

S1 Forward UCED Formulation

The forward model solves independent 168-hour weekly unit-commitment and economic-dispatch (UCED) problems with fixed infrastructure. It co-optimizes commitment, dispatch, zonal exchange, storage and reservoir operation, load shedding, and upward/downward reserves. The formulation below groups the implemented constraints and shows where the inferred parameters enter. Notation follows HOPE where practical; the equations represent `model/EDUCModel.jl`, not additional HOPE model options.

Symbol	Meaning
$h \in H, i \in I, l \in L$	Hours, zones, and transmission corridors.
$G, G^{UC}, G^{CHP}, G^{VRE}, G^{HY}$	Generators; UC, CHP, wind/solar, and hydro subsets.
$s \in S, b \in B, A = G \cup S$	Electrical storage, shedding segments, and all resources.
$p_{g,h}, c_{s,h}, dc_{s,h}$	Generation, storage charging, and discharging (MW).
$o_{g,h}, su_{g,h}, sd_{g,h}$	Binary online, startup, and shutdown indicators.
$soc_{a,h}, q_{g,h}$	Stored energy (MWh) and hydro spillage (MW equivalent).
$f_{l,h}^+, f_{l,h}^-, p_{i,b,h}^{LS}$	Directional flows and unserved demand (MW).
$r_{a,h}^+, r_{a,h}^-, \xi_{l,h}^+, \xi_{l,h}^-$	Reserves and MLT-band relaxation slacks (MW).

Subscripts i on resource sets indicate location. Define $x_{a,h} = p_{a,h}$ for generators and $x_{a,h} = dc_{a,h}$ for storage. With one-hour steps, the operating-cost objective is

$$\begin{aligned}
 \min \quad & \sum_{h \in H} \omega_h \left[\sum_{a \in A} VCG_{a,h} x_{a,h} + \sum_{i,b} VOLL_{i,b} p_{i,b,h}^{LS} \right] \\
 & + \sum_{g \in G^{UC}, h} STC_{g,h} P_g^{unit} su_{g,h} + \pi^{MLT} \sum_{l,h} (\xi_{l,h}^+ + \xi_{l,h}^-). \tag{S1}
 \end{aligned}$$

VCG , $VOLL$, and $STC P^{unit}$ denote dispatch, shedding, and startup costs; P^{unit} is the input unit size. The current weights are $\omega_h = 1$, and $\pi^{MLT} = 10^7$. Capacity auxiliaries in the code are fixed to existing capacities and written here as parameters.

Power balance and shedding. For every zone/hour,

$$\sum_{g \in G_i} p_{g,h} + \sum_{s \in S_i} (dc_{s,h} - c_{s,h}) + \sum_l \Phi_{i,l,h} + \sum_b p_{i,b,h}^{LS} = Load_{i,h}, \quad 0 \leq \sum_b p_{i,b,h}^{LS} \leq Load_{i,h}. \tag{S2}$$

Φ is loss-adjusted net corridor injection. All continuous operating variables are nonnegative. The remaining constraint families follow.

Output, commitment, and ramping.

$$AF_{g,h}\chi_{g,w}\mu_g P_g^{max} o_{g,h} \leq p_{g,h} \leq AF_{g,h} P_g^{max} o_{g,h}, \quad g \in G^{UC}, \quad (S3)$$

$$\mu_a P_a^{max} \leq x_{a,h} \leq AF_{a,h} P_a^{max}, \quad a \in A \setminus G^{UC}, \quad (S4)$$

$$(o_g, su_g, sd_g, p_g) \in \mathcal{U}_g(\mu_g, P_g^{max}, RU_g, RD_g, UT_g, DT_g), \quad g \in G^{UC}. \quad (S5)$$

AF is hourly availability and μ the minimum-output fraction. $\mathcal{U}_g$ collects status transitions, binary bounds, minimum up/down windows, and commitment-adjusted ramp limits. Non-UC resources have ordinary up/down ramp limits. Wind/solar curtailment equals available output minus dispatch. Non-UC minimum output is zero in the intended inputs.

Seasonal CHP. During heating weeks,

$$\sum_{g \in G^{CHP}} P_g^{max} o_{g,h} \geq 0.80 \sum_{g \in G^{CHP}} P_g^{max}. \quad (S6)$$

For CHP, $\chi_{g,w}$ is 1.20 in deep winter, 1.05 in shoulder heating weeks, and 1 otherwise; for non-CHP units it is always 1. Heating weeks are 1–15 and 42–52; deep-winter weeks are 1–8 and 46–52.

Storage and reservoir hydro. At hourly resolution,

$$soc_{s,h} = soc_{s,h-1} + \eta_s^{ch} c_{s,h} - dc_{s,h}/\eta_s^{dis}, \quad 0 \leq soc_{s,h} \leq E_s^{max}, \quad (S7)$$

$$soc_{g,h} = soc_{g,h-1} + I_{g,h} - q_{g,h} - p_{g,h}/\eta_g^{dis}, \quad \beta E_g^{max} \leq soc_{g,h} \leq E_g^{max}. \quad (S8)$$

The first row applies to $s \in S$ and the second to $g \in G^{HY}$. Hydro inflow is $I_{g,h} = AF_{g,h} P_g^{max} \eta_g^{ch}$, with $0 \leq q_{g,h} \leq I_{g,h}$. Initial energy is αE^{max} ; final storage energy may differ by $\pm 0.1 E^{max}$, while hydro returns to its initial level. Storage has a shared charging/discharging/reserve power limit. There is no binary charging/discharging exclusivity. α and β are input fractions.

Transmission and MLT schedules. Set $\kappa_l = 1 - \ell_l/2$ for loss fraction ℓ_l . Directional flows satisfy $0 \leq f_{l,h}^\pm \leq \kappa_l F_l^{max}$. The code uses $\Phi_{i,l,h} = -\sum_{d \in \{+, -\}} [A_{i,l}^d f_{l,h}^d + |A_{i,l}^d| \ell_l f_{l,h}^d / (2\kappa_l)]$, with incidence +1 at the sender and -1 at the receiver. For signed delivery benchmark $m_{l,h}$,

$$(1 - \delta)|m_{l,h}| \kappa_l - \xi_{l,h}^- \leq f_{l,h}^{dir} \leq (1 + \delta)|m_{l,h}| \kappa_l + \xi_{l,h}^+, \quad f_{l,h}^{opp} = 0. \quad (S9)$$

The direction follows the benchmark sign. These schedule restrictions apply in PriorityMLT; SpotOnly omits them. Physical limits and designated one-way external exports remain. MLT slacks provide a numerical/modeling relaxation and diagnostic; they do not define an inference admissibility classifier.

Zonal reserves. Resource-specific sets $\mathcal{R}_a$ collect power headroom, ramping, and storage/hydro energy limits. For every zone/hour,

$$(r_{a,h}^\uparrow, r_{a,h}^\downarrow) \in \mathcal{R}_a(x, o, c, soc), \quad (S10)$$

$$\sum_{a \in A_i} r_{a,h}^\uparrow \geq \lambda_i^D Load_{i,h} + \lambda_i^{VRE} \sum_{g \in G_i \cap G^{VRE}} p_{g,h} + Q_i, \quad (S11)$$

Q_i is the configured largest-unit/interconnection contingency. The inferred vector $\theta = (\delta, \{\mu_k, \tau_k\}_k)$ enters the MLT band, minimum-output bounds, and UC windows ($UT_g = DT_g = \text{round}(\tau_k)$ for units in class k). Other inputs remain fixed within each year's inference.

S2 Candidate Parameterization and Preliminary Screening

S2.1 Preliminary coal-side screening

The preliminary exercise described in the original supplement compared CHP and non-CHP coal parameter responses over a three-month design. CHP minimum-output parameters showed the clearer coal-generation response, especially in the two smaller size classes. Non-CHP parameters showed limited response under that design and were not carried into the full-year inference experiment. This is a design-specific screening result, not a claim that non-CHP flexibility is generally unimportant or intrinsically unrecoverable. No new preliminary-screening estimates are introduced in this revision.

S2.2 Candidate domain and generator grouping

The initial experiment uses a 100-point LHS design over a prespecified 13-dimensional domain: six minimum-output fractions, six common minimum up/down-time controls, and one MLT band. Table S1 lists the fixed domain boundaries, not realized sample minima and maxima.

R denotes official retrofitted coal pilot units, including both CHP and conventional coal where present. **C** denotes non-retrofitted CHP. These groups are crossed with three capacity classes in MW. The time control is sampled numerically and rounded to an integer before being assigned to both minimum up and down times. Other non-retrofitted conventional coal units use fixed minimum output 0.40 and up/down-time inputs of 8 hours in the current input-generation workflow.

Table S1: Complete candidate domain. “min” is a capacity fraction; “time” is in hours. All 13 dimensions belong to the initial design in both years.

ID	Parameter	2016 bounds	2021 bounds	Years
P1	R-min 660-1000	[0.30, 0.75]	[0.30, 0.75]	Both
P2	R-time 660-1000	[4.00, 16.00]	[4.00, 8.00]	Both
P3	R-min 300-660	[0.30, 0.75]	[0.30, 0.75]	Both
P4	R-time 300-660	[4.00, 16.00]	[4.00, 8.00]	Both
P5	R-min 0-300	[0.30, 0.75]	[0.30, 0.75]	Both
P6	R-time 0-300	[4.00, 12.00]	[2.00, 6.00]	Both
P7	C-min 660-1000	[0.30, 0.75]	[0.30, 0.75]	Both
P8	C-time 660-1000	[4.00, 16.00]	[4.00, 16.00]	Both
P9	C-min 300-660	[0.30, 0.75]	[0.30, 0.75]	Both
P10	C-time 300-660	[4.00, 16.00]	[4.00, 16.00]	Both
P11	C-min 0-300	[0.30, 0.75]	[0.30, 0.75]	Both
P12	C-time 0-300	[4.00, 12.00]	[4.00, 12.00]	Both
P13	MLT band	[0.05, 0.40]	[0.05, 0.40]	Both

Retrofitted coal is a policy-defined group, whereas CHP and its seasonal constraints are technology-defined. Legacy exploration files use `CHP_Retrofitted_*` as an alias for `Coal_Retrofitted_*`; the alias does not imply that all retrofitted units are CHP. Only the technology-defined CHP subset receives the seasonal multiplier in S1.

Year-specific time bounds are taken from the frozen `lhs_design_domain.json` files. Their metadata documents reconstruction of the original LHS bounds from stratification, rather than use of realized extrema as new limits. The 2021 retrofitted-unit time bounds differ from 2016; they should not be replaced by the current shared configuration when reproducing the archived experiments.

The 100 evaluations in each year are partitioned into 80 development observations and 20 independent held-out observations. The held-out observations are excluded from GP fitting, cross-validation, and ARD screening.

S3 GP Training, Validation, and ARD Screening

S3.1 Specification and validation

The response is the same quantity mismatch as in the main paper:

$$L(\theta) = \frac{1}{4} \left(E_{coal}^2 + E_{wind}^2 + E_{solar}^2 + E_{exchange}^2 \right). \quad (\text{S12})$$

Generation errors use capacity-normalized regional RMSE, averaged with equal regional weights; exchange uses the observed-flow normalization described in the main paper. No square root is taken after the weighted squared component sum. The saved transformations for both years apply no Box–Cox transformation. Inputs and loss are centered and scaled using training data, with fold-specific preprocessing during cross-validation.

The reported models use zero-mean GPs with ARD Matérn kernels. The final 2016 fit uses Matérn 5/2, while the final 2021 fit uses Matérn 3/2:

$$\begin{aligned} k_{3/2}(z, z') &= \sigma_f^2 (1 + \sqrt{3}r) \exp(-\sqrt{3}r), \\ k_{5/2}(z, z') &= \sigma_f^2 (1 + \sqrt{5}r + 5r^2/3) \exp(-\sqrt{5}r), \\ r^2 &= \sum_j (z_j - z'_j)^2 / \ell_j^2. \end{aligned} \quad (\text{S13})$$

A single fitted nugget σ_n^2 enters the covariance diagonal. It accommodates numerical tolerances and unresolved response variation; it is not an estimated uncertainty distribution over institutional parameters. The kernel family is held fixed in the reported ARD and reduced-GP fits. The saved-model evidence does not require retaining the old supplement’s nested kernel-selection or composite NLL/coverage selection procedure.

Table S2: GP configuration used for the reported fits. Bounds are in natural logs.

Item	Setting
Kernel and mean	Matérn 5/2 ARD (2016) and Matérn 3/2 ARD (2021); zero mean after standardization.
Hyperparameter estimation	Bounded maximum marginal likelihood; no hyperpriors.
Bounds	$\log \ell_j \in [-3, 4]$; $\log \sigma_f \in [-3, 3]$; $\log \sigma_n \in [\log(10^{-3}), \log(0.5)]$.
Three initializations	$\ell_j \in \{0.5, 1, 2\}$, with $\sigma_f = 1$, $\sigma_n = 0.05$.
Validation	Saved independent test partition; five development folds.
ARD criterion	Drop when $\log \ell_j \geq 3.9$ in at least four of five fits.

Table S3: Latest full- and reduced-GP validation. Test predictions were regenerated from each saved model on its frozen held-out rows.

Year	GP	CV R^2	CV RMSE ^a	Test R^2	Test RMSE ^b	n_{test}
2016	Full	0.9982	0.0424	0.9979	1.387	20
2016	Reduced	0.9988	0.0343	0.9987	1.084	20
2021	Full	0.9790	0.1440	0.9936	1.389	20
2021	Reduced	0.9853	0.1204	0.9933	1.419	20

^aCV RMSE uses the saved standardized responses; pooled CV R^2 is reported, not the mean of fold R^2 values. ^bTest RMSE is on the original loss scale, in units of 10^{-4} . Both current test sets contain 20 observations.

The held-out data serve reporting rather than kernel or feature selection. Test R^2 uses the held-out mean as the constant reference. The reduced models achieve test $R^2 = 0.9987$ in 2016 and 0.9933 in 2021, with original-scale RMSE of 1.08×10^{-4} and 1.42×10^{-4} , respectively. Relative to the corresponding full fits, the reduced fit improves the 2016 test metrics slightly and changes the 2021 metrics only marginally: test R^2 changes from 0.9936 to 0.9933 and RMSE from 1.39×10^{-4} to 1.42×10^{-4} . Saved reduced-GP 95% predictive-interval coverage is 90% for 2016 and 100% for 2021.

S3.2 Fold-wise ARD screening criterion

Let $\hat{\ell}_j^{(f)}$ be the length scale estimated using the training portion of development fold f . The fixed criterion is

$$v_j = \sum_{f=1}^5 \mathbf{1}_{\{\log \hat{\ell}_j^{(f)} \geq 4 - 0.1\}}, \quad j \text{ is screened out if } v_j \geq 4. \quad (\text{S14})$$

The complete results are shown below. An upper-bound length scale indicates weak resolved variation of the loss along a dimension over the sampled domain. ARD does not establish parameter recoverability; that question requires the low-mismatch-region diagnostics in S4.

Table S4: Full 13-dimensional ARD screening. F1–F5 are fitted log length scales; hits are counted before rounding. R/C labels follow Table S1.

Year	Parameter	F1	F2	F3	F4	F5	Hits	Rule
2016	R-min 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-min 300-660	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-time 300-660	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-min 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-time 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	C-min 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	C-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	C-min 300-660	2.18	2.06	2.00	2.07	2.06	0/5	Retain
2016	C-time 300-660	4.00	4.00	4.00	3.85	4.00	4/5	Drop
2016	C-min 0-300	2.07	2.11	1.99	2.06	2.17	0/5	Retain
2016	C-time 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	MLT band	1.29	1.40	1.26	1.31	1.23	0/5	Retain
2021	R-min 660-1000	4.00	4.00	4.00	3.41	4.00	4/5	Drop
2021	R-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	R-min 300-660	2.52	2.67	2.49	2.37	2.39	0/5	Retain
2021	R-time 300-660	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	R-min 0-300	2.73	4.00	2.74	4.00	4.00	3/5	Retain
2021	R-time 0-300	4.00	3.13	2.86	4.00	3.02	2/5	Retain
2021	C-min 660-1000	4.00	4.00	4.00	3.92	4.00	5/5	Drop
2021	C-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	C-min 300-660	2.03	2.51	2.49	1.65	2.14	0/5	Retain
2021	C-time 300-660	4.00	4.00	3.62	4.00	4.00	4/5	Drop
2021	C-min 0-300	2.34	2.49	2.50	2.45	2.51	0/5	Retain
2021	C-time 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	MLT band	0.84	1.06	0.99	1.09	1.16	0/5	Retain

S3.3 Reduced GP and retained dimensions

The 2016 baseline screening retains three parameters: MLT band and non-retrofitted CHP minimum output in the 300–660 and 0–300 MW groups. The 2021 baseline retains these three plus retrofitted-coal minimum output in the 300–660 and 0–300 MW groups and retrofitted-coal minimum up/down time in the 0–300 MW group, for six retained dimensions in total.

The reduced GP is refitted using only these retained coordinates from the development design. It approximates the reduced empirical response represented by that design. It is neither the loss conditional on a particular value of the discarded parameters nor an exact analytical marginalization over them. Variation in omitted dimensions can contribute to residual variation.

The 2021 R-min 0–300 dimension reaches the ceiling in three folds and the R-time 0–300 dimension in two folds. Both therefore remain below the specified four-of-five exclusion threshold. A rounded median log length scale alone would not reproduce these fold-wise decisions.

Screening stability and cross-fitted validation

The upper-bound check repeats five-fold screening with *length-scale* ceilings multiplied by 0.5, 1, and 2: $\log$ ceilings $4 + \log(0.5)$, 4, and $4 + \log 2$. The hit tolerance remains 0.1 log units.

Table S5: Ceiling hits under alternative upper bounds. Four or more hits imply exclusion under that setting.

Parameter	2016: 0.5×	1×	2×	2021: 0.5×	1×	2×
R-min 660-1000	5/5	5/5	5/5	5/5	4/5	4/5
R-time 660-1000	5/5	5/5	5/5	5/5	5/5	5/5
R-min 300-660	5/5	5/5	5/5	0/5	0/5	0/5
R-time 300-660	5/5	5/5	5/5	5/5	5/5	5/5
R-min 0-300	5/5	5/5	5/5	3/5	3/5	3/5
R-time 0-300	5/5	5/5	5/5	3/5	2/5	2/5
C-min 660-1000	5/5	5/5	5/5	5/5	5/5	4/5
C-time 660-1000	5/5	5/5	3/5	5/5	5/5	5/5
C-min 300-660	0/5	0/5	0/5	0/5	0/5	0/5
C-time 300-660	5/5	4/5	5/5	5/5	4/5	4/5
C-min 0-300	0/5	0/5	0/5	0/5	0/5	0/5
C-time 0-300	5/5	5/5	5/5	5/5	5/5	4/5
MLT band	0/5	0/5	0/5	0/5	0/5	0/5

The 2021 retain/drop set is unchanged across the three tested ceilings. In 2016, doubling the ceiling changes C-time 660–1000 from excluded to retained: its ceiling-hit count falls from five to three. The three parameters retained by the 2016 baseline remain retained under every tested ceiling, but the complete 2016 retain/drop set is therefore not invariant to the upper-bound specification.

A five-outer-fold check repeats five-fold ARD inside each outer training set and compares full/reduced predictions on the excluded fold, addressing selection leakage in fixed-active-set CV.

Table S6: Cross-fitted full/reduced comparison. RMSE is in standardized response units; the change is relative to the full model.

Year	Full R^2	Reduced R^2	Full RMSE	Reduced RMSE	Δ RMSE
2016	0.9981	0.9988	0.0433	0.0343	-20.74%
2021	0.9790	0.9824	0.1440	0.1321	-8.26%

The cross-fitted reduced models improve aggregate out-of-fold prediction: RMSE decreases by 20.74% in 2016 and 8.26% in 2021. The corresponding R^2 values change from 0.9981 to 0.9988 and from 0.9790 to 0.9824. The archived overall robustness status is FAIL_REVIEW for 2016 because of its ceiling sensitivity and PASS for 2021.

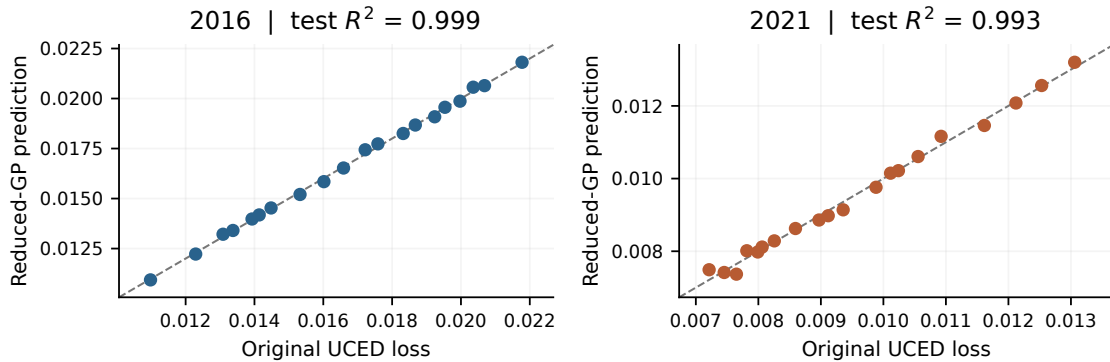


Figure S1: Latest reduced-GP predictions on independent held-out evaluations. Dashed lines indicate equality; no points are removed from the plots.

S4 Low-Mismatch Region and Recoverability Diagnostics

S4.1 Definition and primary region

For each year, the latest saved reduced GP is evaluated on the same stored 20,000-point stratified LHS pool in retained-parameter space (seed 123). The coordinates span the domain in S2; no original UCED solves are added. With $\hat{L}_{min}$ denoting the minimum predicted loss in that pool,

$$S_\epsilon = \{\theta_A \text{ in the evaluated pool} : \hat{L}(\theta_A) \leq (1 + \epsilon)\hat{L}_{min}\}, \quad \epsilon = 0.10. \quad (\text{S15})$$

This finite-pool approximation describes a relative low-mismatch region, not a confidence or credible region. Its reference minimum is not asserted to be a global optimum. For dimension j , concentration is summarized by

$$\text{nIQR}_j(S_\epsilon) = \frac{Q_{0.75}(\theta_j | S_\epsilon) - Q_{0.25}(\theta_j | S_\epsilon)}{U_j - L_j}. \quad (\text{S16})$$

$[L_j, U_j]$ is the corresponding design domain. Lower nIQR indicates greater concentration within this pool and tolerance, not unique identification.

The refreshed pool minima are 0.0104338 in 2016 and 0.0071058 in 2021. The primary regions contain 260 (1.30%) and 3420 (17.10%) of their respective pools. Both stored pools were evaluated with their final saved reduced GP models for a consistent reporting snapshot.

Table S7: Retained-parameter distributions within the latest $S_{10\%}$. All listed parameters are dimensionless. Min/max are finite-pool extrema.

Year	Parameter	Min	Q25	Median	Q75	Max	nIQR
2016	C-min 300-660	0.501	0.633	0.677	0.719	0.749	0.192
2016	C-min 0-300	0.477	0.621	0.669	0.714	0.748	0.208
2016	MLT band	0.050	0.054	0.060	0.068	0.095	0.039
2021	R-min 300-660	0.300	0.446	0.559	0.659	0.750	0.473
2021	R-min 0-300	0.300	0.422	0.536	0.645	0.750	0.495
2021	R-time 0-300	2.001	3.177	4.198	5.112	5.997	0.484
2021	C-min 300-660	0.301	0.436	0.528	0.648	0.750	0.473
2021	C-min 0-300	0.300	0.416	0.530	0.639	0.750	0.495
2021	MLT band	0.050	0.065	0.080	0.098	0.167	0.095

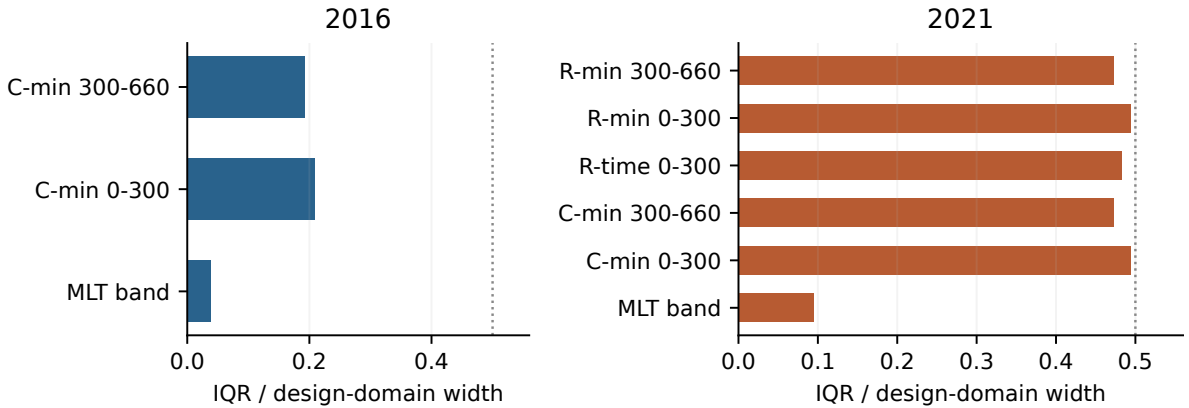


Figure S2: Domain-normalized dispersion in $S_{10\%}$. The dotted 0.5 reference is the IQR of a uniform distribution over the full domain, not a statistical decision threshold.

MLT is more concentrated than the retained coal minimum-output parameters in both years (nIQR 0.039 and 0.095). The additional 2021 retrofitted-coal dimensions have nIQR near 0.5: their ARD relevance does not translate into narrow parameter-level ranges.

S4.2 Tolerance sensitivity and joint relationships

Table S8: Normalized IQR under alternative relative mismatch tolerances.

Year	Parameter	5%	10%	20%	30%
2016	C-min 300-660	0.121	0.192	0.395	0.472
2016	C-min 0-300	0.123	0.208	0.378	0.472
2016	MLT band	0.020	0.039	0.065	0.094
2021	R-min 300-660	0.330	0.473	0.491	0.497
2021	R-min 0-300	0.495	0.495	0.501	0.501
2021	R-time 0-300	0.476	0.484	0.497	0.499
2021	C-min 300-660	0.442	0.473	0.488	0.486
2021	C-min 0-300	0.474	0.495	0.503	0.500
2021	MLT band	0.047	0.095	0.177	0.242

Table S9: Number of pool members in each tolerance region; each pool contains 20,000 points.

Year	5%	10%	20%	30%
2016	53	260	1569	3415
2021	859	3420	7041	9676

The MLT band remains the most concentrated retained dimension at each tested tolerance in both years. Its range broadens as the tolerance increases, especially in 2021. Coal-parameter dispersion is also tolerance-dependent; the narrowest 2016 region has relatively few sampled members and should not be read as a precise interval estimate. All summaries remain conditional on the parameter bounds and the fitted reduced surface.

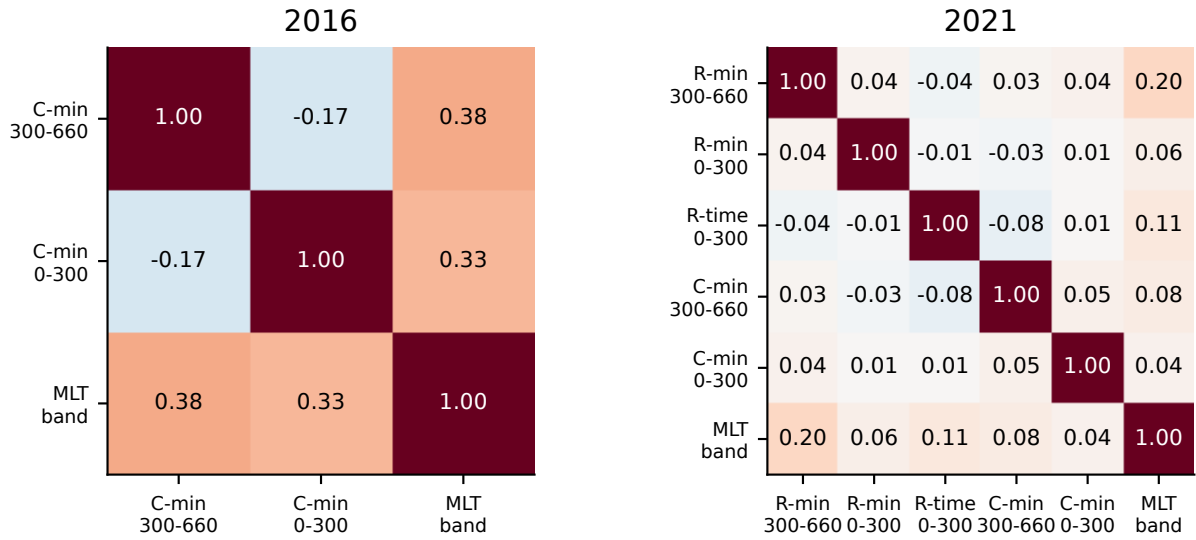


Figure S3: Spearman rank correlations among retained parameters within the latest $S_{10\%}$. These describe the selected region's geometry; they are neither causal effects nor tests of structural identification.

The 2016 region exhibits a clearer negative relationship between the two non-retrofitted CHP minimum-output dimensions than the corresponding 2021 region. Relationships with MLT can reflect compensation between delivery flexibility and minimum-generation rigidity, but pairwise association alone does not establish a unique mechanism. The weak 2021 coal-coal correlations also illustrate that broad parameter ranges need not follow a single narrow tradeoff curve.

Interpretation and scope of the region diagnostics

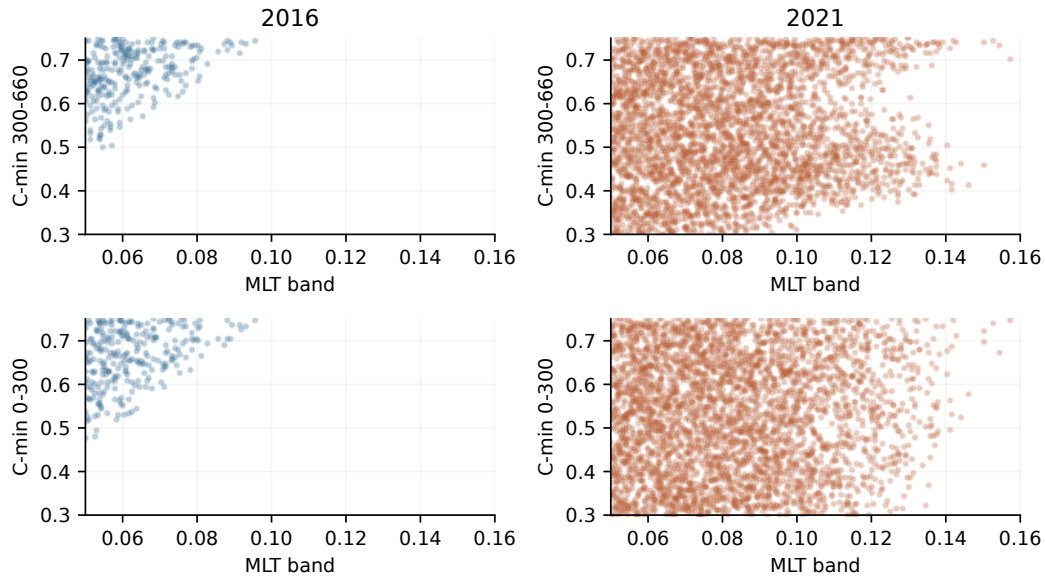


Figure S4: Pairwise projections of $S_{10\%}$ for MLT and the two shared non-retrofitted CHP minimum-output parameters. All region members are plotted; projections conceal variation in other retained dimensions.

The evidence supports different levels of interpretation across parameters. MLT has a relatively concentrated model-ready delivery representation; the coal minimum-output dimensions support a broader diagnosis of operating rigidity. None of these sets uniquely identifies a policy or a causal reform effect. Cross-year differences are interpreted as regime-specific effective UCED representations, conditional on each year's inputs, observations, and parameter domain.

The lower MLT domain boundary is 0.05. Concentration near this edge cannot show whether a still narrower band would also match the quantities, because that region was not evaluated. The 2016 screening sensitivity in S3 and the limited spatial coverage of the six-point-per-year revalidation in S5 constrain how specifically the surrogate region can be interpreted. In particular, GP-UCED residuals at fixed values of omitted parameters can combine approximation error and variation associated with those parameters; they are not a formal upper bound on either component.

S5 Original-UCED Revalidation and Quantity Matching

For each year, six parameter vectors were evaluated using the original UCED: a center representative (labeled “medoid” in the files), four interior points, and a surrogate-best point. Roles refer to the respective archived selection, not a new optimization. GP predictions below use each year’s current reduced model at the exact evaluated coordinates; this accounts for rounding continuous candidate parameters to three decimals before UCED runs. All twelve parameter vectors match their archived candidate records.

S5.1 2016 revalidation

The center representative is nearest the region centroid in domain-normalized coordinates, and the four interior points span distance quartiles within the region. The surrogate-best point is the lowest-predicted-loss pool member. All six satisfy the final 2016 $S_{10\%}$ cutoff, approximately 0.011477. Omitted parameters remain fixed at their prescribed midpoint values.

Table S10: 2016 original-UCED revalidation. Losses are on the original scale; residuals are UCED minus GP.

Point	GP loss	UCED loss	UCED – GP
Center representative	0.011201	0.011088	-0.000112
Interior 1	0.010955	0.010754	-0.000201
Interior 2	0.011148	0.010922	-0.000225
Interior 3	0.011286	0.011151	-0.000136
Interior 4	0.011442	0.011189	-0.000253
Surrogate best	0.010423	0.010087	-0.000335

The six UCED losses range from 0.010087 to 0.011189. The current GP overpredicts loss at all six points, with a mean absolute residual of 0.000210 and a maximum of 0.000335. The median absolute relative discrepancy is 1.97%. The surrogate-best point has the lowest UCED loss among these six; the center and four interior points range from 0.010754 to 0.011189. Thus the evaluations provide evidence of comparable matching at several selected representations, while also revealing local surrogate error. They do not establish uniform accuracy across the region.

Table S11: 2016 component NRMSE values. Aggregate loss is one quarter of the sum of the four squared components, using unrounded values.

Point	Coal	Wind	Solar	Exchange
Center representative	0.0840	0.1297	0.1214	0.0757
Interior 1	0.0832	0.1275	0.1193	0.0750
Interior 2	0.0838	0.1274	0.1190	0.0792
Interior 3	0.0844	0.1304	0.1221	0.0746
Interior 4	0.0838	0.1321	0.1231	0.0715
Surrogate best	0.0816	0.1242	0.1167	0.0681

Coal NRMSE ranges from 0.0816 to 0.0844, wind from 0.1242 to 0.1321, solar from 0.1167 to 0.1231, and exchange from 0.0681 to 0.0792. The aggregate agreement should therefore be read together with component errors rather than as evidence of equally close matching for every quantity.

S5.2 2021 revalidation

The 2021 center representative is nearest the region centroid in domain-normalized coordinates, and the four interior points are sampled from distance quartiles. The surrogate-best point is the lowest-predicted-loss pool point and provides an additional minimum-selection diagnostic. All six satisfy its 2021 $S_{10\%}$ loss cutoff, approximately 0.007816. Omitted parameters remain fixed at their prescribed midpoint values.

Table S12: 2021 original-UCED revalidation. Losses are on the original scale; residuals are UCED minus GP. Candidate roles refer to the pool-based selection before parameter rounding.

Point	GP loss	UCED loss	UCED – GP
Center representative	0.007503	0.007161	-0.000342
Interior 1	0.007702	0.007371	-0.000331
Interior 2	0.007335	0.007168	-0.000168
Interior 3	0.007767	0.007309	-0.000458
Interior 4	0.007684	0.007624	-0.000059
Surrogate best	0.007107	0.007065	-0.000042

The six original-UCED losses range from 0.007065 to 0.007624. The GP overpredicts loss at all six points, with a mean absolute residual of 0.000233 and a maximum of 0.000458. The median absolute relative discrepancy is 3.41%. These are descriptive checks at selected points, not an independent estimate of region-wide prediction accuracy. The surrogate-best point yields the lowest evaluated loss among these six, 0.007065, and its residual magnitude is about 0.59% of its GP prediction. Several distinct parameter vectors retain similar quantity mismatch under the original model.

Table S13: 2021 component NRMSE values. The aggregate loss equals one quarter of the sum of their squares, using unrounded values.

Point	Coal	Wind	Solar	Exchange
Center representative	0.1232	0.0828	0.0362	0.0727
Interior 1	0.1230	0.0829	0.0360	0.0787
Interior 2	0.1241	0.0830	0.0362	0.0712
Interior 3	0.1229	0.0829	0.0362	0.0771
Interior 4	0.1255	0.0830	0.0350	0.0815
Surrogate best	0.1233	0.0829	0.0364	0.0697

Coal, wind, and solar errors vary little across these points; exchange errors show a wider range (0.0697–0.0815). The component results support similar aggregate matching at the selected representations, without establishing uniform accuracy across all hours, regions, or all of $S_{10\%}$. The checks support the use of a low-mismatch region rather than identifying one uniquely preferred institutional parameterization. They do not resolve the screening sensitivity or parameter-domain limitations discussed above.

S6 Reproducibility and Repository Contents

The package includes LaTeX, tables, figures, plotting data, and a SHA-256 source manifest. Scripts regenerate held-out and dense-pool predictions from saved GPs; they do not refit models or run UCED. The repository manifest records Julia 1.11.3 and GaussianProcesses.jl 0.12.6. Frozen splits, folds, domains, and model files preserve the run-specific settings.

Formulation references. HOPE PCM and notation; M. R. Davidson and J. I. Pérez-Arriaga (2018), “Modeling Unit Commitment in Political Context,” *IEEE Transactions on Power Systems*, 33(5), 4889–4901.